

The Clean Two-Query Threshold for Unitary Synthesis

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Abstract

Aaronson and Kuperberg asked whether every n -qubit unitary can be implemented by a polynomial-size quantum circuit making few queries to a classical oracle. At one query the answer is known to be no. We study two sequential queries, allowing arbitrary workspace but requiring its approximate return to a fixed state.

We settle the two-query case in the clean model, where the workspace must be returned to its initial state, by pinning both of its resource thresholds up to a logarithm in the error. Write $N = 2^n$.

- **Width.** For any fixed two-query architecture $T_{x,y} = B D_x W D_y C$ with arbitrary contractions B, W, C and independent unimodular phase selectors x, y , the Gaussian mean width of the realizable family is at most $2 \min(N, Q) \sqrt{\log 2Q}$, where Q is the dimension of the register the oracle acts on, ancillas included. The supremum over selectors is absorbed pointwise; no chaining is used.
- **Lower bounds.** Consequently a universal clean two-query circuit needs $\Omega(2^n)$ qubits, and, by an address-compression lemma, an oracle input length of $\Omega(2^n)$ bits regardless of workspace.
- **Matching upper bound.** A permutation-transport construction implements every unitary cleanly with two queries, $O(2^n \log(1/\varepsilon))$ qubits and $O(2^n \log(1/\varepsilon))$ address bits, at error ε .

So the clean two-query qubit threshold and address-length threshold are both $\Theta(2^n)$ up to the factor $\log(1/\varepsilon)$. We also show that the width bound is tight at register dimension $Q = \Theta(N^2)$, explain why the one-query argument does not extend, and prove a transfer theorem reducing the garbage-model two-query problem to a clean four-insertion width bound, which remains open. This is a sufficient route, not a proof that every garbage-model method must use four insertions.

Verification note. Selected identities and finite instances are exercised by scripts in `verify/` of the [accompanying repository](#), and `STATUS.md` there records, claim by claim, which checks exist, what has been independently audited and what has not. These checks do not certify the complete analytic proof. File names in this paper are paths in that repository.

1 Introduction

1.1 The problem

The unitary synthesis problem of Aaronson and Kuperberg [AK07] asks whether every n -qubit unitary can be implemented by a polynomial-time quantum algorithm with few queries to a classical oracle. The known landmarks are:

- [AK07, Thm 6.7] one query, exact model: at most 4^N distinct unitaries, at most $2N$ relevant Boolean table entries (not address bits);

- [LMW24] one query, or polynomially many parallel (non-adaptive) queries, in the approximate/discard model, oracle input length $o(2^n)$;
- [DLM26] explicit one-query separations, and Question 1.4 posing the general-depth problem;
- [BY26] an oracle input-length bound $(2 - o(1)) \log d$ valid at any query count;
- [Ros26] an upper bound: $O(2^{n/2})$ clean queries suffice, with $\text{poly}(n)$ ancillas.

This paper proves a workspace-uniform capacity bound for two sequential queries in the clean model and a matching fixed-error resource threshold. Related restricted work includes Banerjee’s exact every-oracle-clean counting analysis [Ban25] and Huang’s 1.5-query study [Hua25]. Those models and statements should not be identified with the present approximation-and-address theorem; none of our proofs uses them.

1.2 The model and the scope of the clean restriction

A fixed circuit with two oracle calls realizes a target U if some oracle makes it implement U on the target register and, in the **clean** model, return every auxiliary register to a fixed state. Nothing is required of the circuit for other oracles; in particular, unlike the setting of [AK07, Thm 6.7], whose footnote 13 notes that the theorem needs the circuit to implement some unitary for every oracle, we impose no such requirement, and our lower bounds hold without it. The Bernstein–Vazirani one-query universal synthesizer, a lookup attributed to Yuen in [LMW24], is not clean: its workspace ends holding the description string. So the class we bound is strictly smaller than the discard model of [LMW24], and **our conclusions are correspondingly stronger on a correspondingly smaller class; they neither improve nor contradict [LMW24] or [BY26]**. Section 7 prices the distance between the two models.

1.3 What is new

[LMW24, Def. B.1] already name the quantity we bound: an oracle circuit is S -small if it realizes at most S distinct unitaries, and their Appendix B proves that small capacity implies a lower bound. They then observe that counting arguments “become ineffective once the adversary has even a small number of ancilla qubits.” The implication is theirs; the missing premise is that a two-query architecture’s capacity *is* small. This paper supplies that premise, in a metric-entropy form that degrades only as the square root of the total qubit count: an adversary needs $\Omega(2^n)$ ancilla qubits to neutralize it.

Two rounds, not two queries. A phase selector D_x is an arbitrary diagonal on the whole register, and a product of phase oracles applied in one round is again such a diagonal. One selector factor therefore subsumes arbitrarily many parallel queries, so Theorem 4.1 constrains *two rounds of adaptivity with arbitrarily many parallel queries per round*.

The obstruction [LMW24, §2.5] name for going past one query, bounding the operator norm of 2^M matrices simultaneously, is met by absorbing the entire selector supremum *pointwise* into a single operator norm (Section 4).

Beyond the width bound, this version adds: the extension from real sign words to independent unimodular selectors, which covers diagonal-unitary oracles and two different tables; an address-length lower bound that holds with any amount of workspace; the matching upper bound; and the garbage-model transfer.

Indexing. [DLM26, Question 1.4] ask whether synthesizing $F_t H \cdots F_1$ requires t sequential queries; the case t is a lower bound against $t - 1$ queries, so their proved case $t = 2$ is a one-query bound and a two-query bound has the strength of their case $t = 3$. Our result has that strength for Haar-scale targets; we say nothing about their explicit family.

2 Setting

Definition 2.1 (two-query architecture). A triple (B, W, C) with C a $Q \times N$ contraction, W a $Q \times Q$ contraction and B an $N \times Q$ contraction, realizing

$$T_{x,y} = B D_x W D_y C, \quad D_x = \text{diag}(x), \quad D_y = \text{diag}(y), \quad x, y \in \mathbb{T}^Q,$$

where $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$. The **repeated real sign word** is the special case $x = y = g \in \{\pm 1\}^Q$, written T_g ; it is the shape of a two-query circuit with one ± 1 phase oracle. Independent x, y cover diagonal-unitary oracles and circuits that query two different tables.

Ancillas are inside Q : an architecture on n target qubits and a ancillas has $Q = 2^{n+a}$, and we write $w = \log_2 Q$ for the total qubit count. The bounds below are stated in $\log(2Q)$, i.e. in w ; Theorem 6.3 then separates the oracle input length from the workspace.

Notation. $b_p = B e_p \in \mathbb{C}^N$ is the p -th column of B and $c_q = e_q^\top C \in \mathbb{C}^N$ the q -th row of C , with **leverages** $\lambda_p = \|b_p\|^2$, $\mu_q = \|c_q\|^2$, $\Lambda = \text{diag}(\lambda)$, $M_\mu = \text{diag}(\mu)$, and P_λ, P_μ the projections onto their supports. Since B and C are contractions, $\sum_p \lambda_p = \|B\|_F^2 \leq \min(N, Q)$ and $\sum_q \mu_q = \|C\|_F^2 \leq \min(N, Q)$.

Proposition 2.2 (unitarity criterion). Write $\Psi(x, y) = D_x W D_y C$ and $\Pi = I_Q - B^* B$. Then

$$T_{x,y}^* T_{x,y} = I_N - (I_N - C^* C) - C^* D_y^* (I_Q - W^* W) D_y C - \Psi(x, y)^* \Pi \Psi(x, y).$$

All three subtracted terms are positive semidefinite, so $T_{x,y}$ is unitary exactly when C is an isometry, $(I_Q - W^* W) D_y C = 0$ and $\Pi \Psi(x, y) = 0$. When C and W are isometries the criterion reduces to $\Pi \Psi = 0$.

Proof. $T^* T = \Psi^* B^* B \Psi = \Psi^* \Psi - \Psi^* \Pi \Psi$ and $\Psi^* \Psi = C^* D_y^* W^* W D_y C = C^* C - C^* D_y^* (I - W^* W) D_y C$, while $C^* C = I_N - (I_N - C^* C)$. Each subtracted term is positive semidefinite because $I - C^* C$, $I - W^* W$ and Π are, for contractions; and a sum of positive semidefinite operators vanishes only when each does. $\square$

Erratum. An earlier draft stated the criterion with only the last term, which is correct only for isometric C and W ; with $C = 0$, a legal contraction, the shorter criterion is satisfied while $T = 0$. `verify_complex_phase.py` (e) checks the three-term identity for contractive C and W and exhibits the failure of the shorter forms.

Definition 2.3 (capacity). The capacity at scale ρ is $\log \text{Pack}_{\rho\sqrt{N}}\{T_{x,y}\}$, the logarithm of the largest family of realizable operators that is pairwise $\rho\sqrt{N}$ -separated in Frobenius norm. This is the physically meaningful scale for unitary targets, since $\|U\|_F = \sqrt{N}$. **The capacity claim is asserted only in this metric**; a raw, un-normalized reading is false by a factor N (Corollary 5.1).

3 Why the one-query argument does not extend

[AK07, Thm 6.7] writes $U(X_{ij}) = I + E_i + E_j$ and applies [AK07, Lem. 6.6] to the anticommutation relations $E_i E_j^\dagger + E_j E_i^\dagger = 0$. This is pure affineness and is unavailable at degree two. One might still hope that a combinatorial invariant of the **lawful set**, the set of selector words for which the circuit's output is unitary, controls capacity. Two facts speak against building on that hope, one at depth two and one beyond it.

Proposition 3.1 (at depth two the final layer is invisible). *For a two-query architecture, write $\Psi(x, y) = D_x W D_y C$ and let $\mathcal{L}$ be its lawful set. Then the number of distinct realizable targets $\{T_{x,y} : (x, y) \in \mathcal{L}\}$ equals the number of distinct $\Psi(x, y), (x, y) \in \mathcal{L}$. Given Ψ and $\mathcal{L}$, the count does not depend on B .*

Proof. For lawful (x, y) , $T = B\Psi$ is unitary, so by Proposition 2.2 Ψ is an isometry and B is isometric on $\text{ran } \Psi(x, y)$. For a contraction B the set of vectors on which it is isometric is the subspace $\ker(I - B^*B)$, so B is injective on the sum of these ranges over the lawful set, and $B\Psi(x, y) = B\Psi(x', y')$ forces $\Psi(x, y) = \Psi(x', y')$. $\square$

Proposition 3.2 (beyond depth two, pre-final data does not determine the count). *In the three-insertion class $\Psi_g = D_g W_2 D_g W_1 D_g C$, $T_g = B\Psi_g$, with $Q = 8$ and a one-dimensional target ($N = 1$), there are two architectures sharing C , W_1 and B and differing only in W_2 , with identical lawful sets (a linear code of 16 sign words closed under pointwise product) and identical pre-final data (the frames C , $D_g C$, $D_g W_1 D_g C$ on the lawful set and their Gram kernels), whose numbers of distinct realizable targets are 2 and 16.*

An explicit rational construction and its exhaustive exact certificate are given in Appendix B. The numerical script `verify_decisive_pair.py` provides supplementary checks, including Proposition 3.1. Since $N = 1$, this example distinguishes scalar matrices, not quantum channels.

Remark 3.3 (what this motivates, and nothing more). Neither fact is a bound. The first says that at depth two capacity is a property of the pre-final map restricted to the lawful set, an object whose Boolean structure we have no tool to count. The second shows that frames ending before the last variable mixer need not determine the final matrices; it does not contradict injectivity of a fixed final contraction on its lawful ranges. Both point away from combinatorial invariants of the lawful set and toward a quantity that sees the operators $T_{x,y}$ themselves: their metric entropy, which Section 4 controls through a Gaussian mean width. A 2-versus-16 gap at $Q = 8$ establishes nothing asymptotically.

4 The width theorem

Terminology. “Width” means the Gaussian mean width of a set of matrices, in the sense of convex geometry. [LMW24, Lem. 4.11] use “width” for an unrelated leverage-normalized overlap statistic.

Let G be an $N \times N$ matrix with independent standard complex Gaussian entries, $\mathbb{E}|G_{ij}|^2 = 1$.

Theorem 4.1 (two-query Gaussian mean width, unimodular selectors). *For every two-query architecture (B, W, C) of Definition 2.1,*

$$w_2 := \mathbb{E}_G \sup_{x,y \in \mathbb{T}^Q} |\text{Tr}(G^* T_{x,y})| \leq \|B\|_F \|C\|_F \cdot 2\sqrt{\log(2Q)} \leq 2 \min(N, Q) \sqrt{\log(2Q)}.$$

The supremum sits inside the expectation; no exchange of supremum and expectation occurs.

Proof. If either endpoint is zero, the assertion is immediate. Otherwise both leverage supports are nonempty. The following six steps keep the selector supremum inside the expectation; the absorption inequality is pointwise.

1. **Exact expansion.** Since $D_x = \sum_p x_p e_p e_p^\top$ and likewise for D_y ,

$$\text{Tr}(G^* T_{x,y}) = \sum_{p,q} x_p y_q W_{pq} (c_q G^* b_p) = x^\top M y, \quad M_{pq} := W_{pq} (c_q G^* b_p).$$

The matrix M does not depend on x or y . If $\lambda_p = 0$ then $b_p = 0$ and row p of M vanishes; if $\mu_q = 0$ then column q vanishes. Restricting to the supports is therefore exact.

2. **Leverage normalization.** Put $K := \Lambda^{-1/2} M M_\mu^{-1/2}$ on the supports and $u = \Lambda^{1/2} x$, $v = M_\mu^{1/2} y$, so that $x^\top M y = u^\top K v$.
3. **Pointwise absorption.** For unimodular selectors the norms $\|u\|^2 = \sum_p \lambda_p |x_p|^2 = \|B\|_F^2$ and $\|v\|^2 = \|C\|_F^2$ are **constant over the torus**. Hence, pointwise in G ,

$$\sup_{x,y \in \mathbb{T}^Q} |x^\top M y| \leq \|B\|_F \|C\|_F \|K\|_{\text{op}}.$$

The supremum over the entire selector torus has been absorbed into one operator norm: no net, no chaining, no union bound. This is the heart of the proof.

Provenance. The inequality of this step is not new. It is the diagonal-scaling relaxation of quadratic maximization over the cube and the torus that runs from Delorme–Poljak and Poljak–Rendl through Nesterov, Alon–Naor and Charikar–Wirth [DP93, PR95, Nes98, AN06, CW04]; the degree- t structure is the polynomial method [BBCMW98], of which [AK07, Thm 6.7] is the $t = 1$ case. What is ours is the composition: polynomial method, then this relaxation, then matrix concentration, then Sudakov, run against a synthesis architecture, and the variance computation of the next step that makes the composition close.

4. **K is a matrix Gaussian series.** Writing $G = \sum_{ij} G_{ij} e_i e_j^\top$,

$$K = \sum_{i,j} \overline{G_{ij}} A_{ij}, \quad (A_{ij})_{pq} = \frac{W_{pq} (b_p)_i (c_q)_j}{\sqrt{\lambda_p \mu_q}}.$$

Its two variance operators have closed forms. With $\Gamma_B := \Lambda^{-1/2} B^* B \Lambda^{-1/2}$ and $\Gamma_C := M_\mu^{-1/2} C C^* M_\mu^{-1/2}$, both positive semidefinite with unit diagonal on their supports,

$$\sum_{ij} A_{ij} A_{ij}^* = \overline{\Gamma_B} \circ (W P_\mu W^*), \quad \sum_{ij} A_{ij}^* A_{ij} = \overline{\Gamma_C} \circ (W^* P_\lambda W),$$

where $\circ$ is the entrywise product. Both closed forms are checked against Monte Carlo in `verify_complex_phase.py` (c).

5. **Both variance operators are dominated by the identity.** If $\Gamma \succeq 0$ has $\Gamma_{pp} \leq 1$ then $\|\Gamma \circ X\|_{\text{op}} \leq \|X\|_{\text{op}}$ for every X : writing $\Gamma_{pp'} = u_p^* u_{p'}$ with $\|u_p\| \leq 1$, the matrix $\Gamma \circ X$ is the compression of $I \otimes X$ by the contraction $e_p \mapsto u_p \otimes e_p$; $\overline{\Gamma}$ is positive semidefinite with the same diagonal, so the lemma applies to it as well. Since $W P_\mu W^*$ and $W^* P_\lambda W$ are positive semidefinite of norm at most one, both variance operators are positive semidefinite of norm at most one, and they are supported on P_λ and P_μ respectively. Only $\|W\| \leq 1$ is used; the middle layer is otherwise arbitrary.
6. **Matrix Khintchine.** For a real Gaussian series $Z = \sum_k \gamma_k A_k$ with $d_1 \times d_2$ coefficients, Tropp's bound gives $\mathbb{E}\|Z\| \leq \sqrt{2v \log(d_1 + d_2)}$ with $v = \max\{\|\sum_k A_k A_k^*\|, \|\sum_k A_k^* A_k\|\}$ [Tro15, Thm 4.1.1]. Splitting each complex Gaussian into real and imaginary parts of

variance $1/2$ gives two real series with variance parameter $v/2 \leq 1/2$, hence $\mathbb{E}\|K\| \leq 2\sqrt{\log(2Q)}$ by the triangle inequality. (Treating the two parts as one real series of $2N^2$ terms gives the sharper $\sqrt{2}$, which is what the verifier tests; the constant 2 is all that is needed.) Assemble with step 3. □

Corollary 4.2 (real sign words; the certified form). *For the repeated real sign word,*

$$\mathbb{E}_G \sup_{g \in \{\pm 1\}^Q} \operatorname{Re} \operatorname{Tr}(G^* T_g) \leq 2 \min(N, Q) \sqrt{\log(2Q)}.$$

Proof. Take $x = y = g$ and use $|\operatorname{Re} z| \leq |z|$. □

The form of this corollary that was independently audited (see `STATUS.md`) carries the constant 8 and is proved by a symmetric weighting: with $d_p = \lambda_p + \mu_p$ and $D = \operatorname{diag}(d)$ one writes $\operatorname{Re}(g^\top M g) = v^* K v$ with $v = D^{1/2} g$, $\|v\|^2 = \operatorname{Tr} D \leq 2 \min(N, Q)$, and bounds $\mathbb{E}\|K\| \leq 4\sqrt{\log 2Q}$. The asymmetric route above was also verified in that audit, as the product-form strengthening with $\|B\|_F \|C\|_F$ in place of $\operatorname{Tr} D$; both routes are valid and incomparable instance by instance. The deterministic spine of the symmetric route is formalized in the `lean/` directory of the repository.

Remark 4.3 (the weighting is sufficient, not unique). Under $d \mapsto td$ the variance operators scale by t^{-2} , so any $d' \succeq \lambda + \mu$ works subject to the trace budget. Balance is necessary: over random contractive architectures, $d = \lambda$ alone or $d = \mu$ alone fails the domination on a substantial fraction of instances, while the balanced or the asymmetric normalization succeeds on all of them.

Remark 4.4 (zero leverage is lossless). $\lambda_p = 0$ kills row p of M and $\mu_q = 0$ kills column q , so the supports may differ and K is rectangular in general. Nothing downstream changes; the concentration step stays at dimension at most $2Q$. The same holds when the middle layer is itself rectangular, $Q_1 \times Q_2$, with a selector of length Q_1 on the left and Q_2 on the right: nothing in steps 1 to 6 uses $Q_1 = Q_2$, the endpoint norms are as before, and the Khintchine dimension becomes $Q_1 + Q_2$.

Remark 4.5 (provenance of the logarithm). $\sqrt{\log(2Q)}$ is the standard Khintchine logarithm, $2Q$ being Tropp's $d_1 + d_2$. It is the same factor appearing as $\sqrt{2 \ln 2M}$ in [LMW24, Lem. 4.11]. Inherited, not discovered.

Remark 4.6 (the selector need not be real). An earlier draft proved only the real case and disclaimed any extension to diagonal-unitary oracles, because its symmetric route rewrote $\operatorname{Re}(g^\top M g)$ as a Hermitian form, which uses $g \in \mathbb{R}^Q$. The asymmetric route never does this: step 3 uses only that $|x_p| = |y_q| = 1$. The real restriction is gone, and with it the restriction to a single table.

5 From width to capacity, and the constant

Corollary 5.1 (capacity). *For fixed $\rho > 0$ and $Q \geq N$,*

$$\log \operatorname{Pack}_{\rho\sqrt{N}}\{T_{x,y}\} \leq \frac{2C_S^2 w_2^2}{\rho^2 N} \leq \frac{\kappa}{\rho^2} N \log(2Q), \quad \kappa := 8C_S^2,$$

where C_S is the constant in Sudakov's minoration for the process $X_T = \operatorname{Re} \operatorname{Tr}(G^* T)$, whose intrinsic distance is $\|S - T\|_F / \sqrt{2}$.

Proof. Sudakov’s minoration [Ver18, Thm 7.4.1], [Tal14]: a family that is ϵ -separated in the process metric and lies in the index set of a Gaussian process with $\mathbb{E} \sup X_T = w$ has $\epsilon \sqrt{\log \text{Pack}} \leq C_S w$. At Frobenius separation $\rho \sqrt{N}$ the process separation is $\rho \sqrt{N/2}$, and

$$\mathbb{E} \sup_T \text{Re Tr}(G^* T) \leq w_2,$$

so inserting Theorem 4.1 gives the bound. The hypothesis $Q \geq N$ always holds, because the target register is part of the register the oracle acts on. $\square$

Erratum, stated where a reader will quote it. Corollary 5.1 is **false in raw Frobenius**, by a factor N . It is true only in the normalized metric of Definition 2.3. This is a corollary, not a link in the proof of Theorem 4.1, but it is the statement downstream consumers quote. Re-check the separation metric before instantiating it.

Corollary 5.2 (qubit lower bound). *For every $0 < \rho < \sqrt{2}$, $\log \text{Pack}_{\rho \sqrt{N}} U(N) \geq c(1 - \rho^2/2)^2 N^2$ for an absolute $c > 0$ and all $N \geq 3$. Hence no fixed clean two-query architecture realizes every unitary, exactly or with operator-norm error below $\rho/4$, unless $\log Q = \Omega_\rho(N)$. In qubit count:*

$$w = \Omega(2^n)$$

for a universal clean two-query architecture on n target qubits.

Proof. For fixed U and Haar-random V , $\|U - V\|_F^2 = 2N - 2 \text{Re Tr}(U^* V)$, so V lies in the Frobenius ball of radius $\rho \sqrt{N}$ about U exactly when $\text{Re Tr}(U^* V) \geq N(1 - \rho^2/2)$. The function $V \mapsto \text{Re Tr}(U^* V)$ has Haar mean zero and is $\sqrt{N}$ -Lipschitz in the Frobenius metric, and concentration of Lipschitz functions on $U(N)$ [Mec19, Thm 5.17] gives $\Pr(f \geq \mathbb{E}f + t) \leq \exp(-c' N t^2 / L^2)$; at $t = N(1 - \rho^2/2)$, $L = \sqrt{N}$, the ball has Haar measure at most $\exp(-c'(1 - \rho^2/2)^2 N^2)$. A maximal $\rho \sqrt{N}$ -separated set is a $\rho \sqrt{N}$ -covering, so its size is at least the reciprocal of that measure. If an architecture realizes each member U_i of such a set to operator-norm error below $\rho/4$, the realized operators are within Frobenius distance $\rho \sqrt{N}/4$ of the U_i and hence $\rho \sqrt{N}/2$ -separated; Corollary 5.1 at scale $\rho/2$ gives $c(1 - \rho^2/2)^2 N^2 \leq 4\kappa \rho^{-2} N \log(2Q)$, i.e. $\log Q = \Omega_\rho(N)$. $\square$

For comparison, the trivial clean two-query lookup, which stores every entry of U in the table, uses $\tilde{\Theta}(N^2)$ qubits, so at this point the threshold sits in $[\Omega(N), \tilde{O}(N^2)]$; Theorem 6.4 closes the gap.

Proposition 5.3 (the constant matters). *The permutation family of Section 6.1 forces $\kappa \geq \frac{1}{4}$ asymptotically, the maximum being approached at separation $\rho^2 = 1$.*

Computed from Gilbert–Varshamov packings of permutation codes, with $d = N$, in the script `verify_crho_requirement.py`:

d	16	256	4096	65536	$\rightarrow \infty$
required κ	0.126	0.159	0.185	0.200	0.25

Sudakov’s minoration supplies a universal finite constant C_S ; we do not claim a numerical upper bound for it. The threshold and tightness results use only a universal constant. The permutation comparison remains a useful normalization check: any valid choice of κ in Corollary 5.1 must obey the displayed lower requirement.

6 Tightness, address length, and the matching upper bound

6.1 Tight at register dimension of order N^2

Proposition 6.1. *The clean two-query permutation-synthesis algorithm of [DLM26] lies inside Definition 2.1 and attains the ceiling of Corollary 5.1 up to the constant factor 4κ .*

[DLM26] synthesize any permutation unitary P_π on $d = N$ dimensions in two clean queries, querying π then π^{-1} . Index addresses by (b, x, y) with $b \in \{0, 1\}$ and $x, y \in [d]$, so $Q = 2d^2$:

$$g_{0,x,y} = (-1)^{y \cdot \pi(x)}, \quad g_{1,x,y} = (-1)^{x \cdot \pi^{-1}(y)},$$

$$C : |x\rangle \mapsto d^{-1/2} \sum_y |0, x, y\rangle, \quad W = X_b \otimes H_X \otimes H_Y, \quad B : |b, x, y\rangle \mapsto \delta_{b,1} d^{-1/2} |y\rangle.$$

W is fixed and independent of π ; a single sign word serves both queries because the state occupies block 0 at the first query and block 1 at the second. Tracing through, $T_g = P_\pi$ exactly. Verified in `verify_permutation_family.py` over all 24 permutations at $d = 4$ and 200 random permutations at $d = 8$ with residuals below 10^{-14} . Since $\|P_\pi - P_\sigma\|_F^2 = 2 \text{Ham}(\pi, \sigma)$, permutation codes at Hamming distance αd give $\rho^2 = 2\alpha$, and Gilbert–Varshamov gives $\log \text{Pack} = \Theta(d \log d) = \Theta(N \log 2Q)$.

Corollary 6.2 (tight in both factors at $Q = \Theta(N^2)$). *Running Sudakov backwards on the permutation family gives $w_2 \geq c\rho\sqrt{d}\sqrt{d \log d} = \Theta(N\sqrt{\log 2Q})$ at $Q = 2N^2$. So at that scale Theorem 4.1 is attained up to a constant, and neither the factor N nor the factor $\sqrt{\log 2Q}$ can be improved by more than a constant there.*

Padding an architecture with unused workspace raises $\log Q$ without changing w_2 , so this is a statement about the scale $Q = \Theta(N^2)$, not about every polynomial. At exponential Q the logarithm is necessary as well: by Theorem 6.4, at $\log Q = \Theta(N \log(1/\varepsilon))$ the class ε -covers $U(N)$ in operator norm, and since $|\text{Tr}(G^*(T - U))| \leq \|G\|_{S_1} \|T - U\|_{\text{op}}$ this gives $w_2 \geq (1 - \varepsilon) \mathbb{E} \|G\|_{S_1} = \Theta(N^{3/2})$ there. No bound of the form $N \cdot \text{polylog}(N)$ can therefore hold uniformly in Q .

6.2 The oracle’s input length alone

Theorem 4.1 bounds the total register dimension. The next theorem removes the workspace from the accounting entirely. In this section and the next, M denotes a number of addresses; the coefficient matrix M of Section 4 does not appear.

Theorem 6.3 (address-length lower bound). *Let the oracle act on the Q -dimensional register through M addresses of arbitrary rank,*

$$D_x = \sum_{j=1}^M x_j P_j, \quad x \in \mathbb{T}^M,$$

with $P_1, \dots, P_M$ mutually orthogonal projections summing to I_Q (for a Boolean table on ℓ input bits, $M = 2^\ell$ and P_j projects onto address j tensored with the whole workspace). Then for every two-query architecture

$$\mathbb{E}_G \sup_{x,y \in \mathbb{T}^M} |\text{Tr}(G^* B D_x W D_y C)| \leq 2 \|B\|_F \|C\|_F \sqrt{\log(2MN)} \leq 2N \sqrt{\log(2MN)},$$

and consequently universal clean two-query synthesis, exact or to operator-norm error below an absolute constant, requires $\log M = \Omega(N)$, i.e. an oracle input length of $\Omega(2^n)$ bits, regardless of the number of ancillas.

Proof. Let $L := \sum_j \text{ran}(P_j B^*)$ and $R := \sum_j \text{ran}(P_j C)$, subspaces of $\mathbb{C}^Q$ of dimension at most MN each. Because the P_j are mutually orthogonal, L is the orthogonal direct sum of the $\text{ran}(P_j B^*)$ and is invariant under every P_j ; likewise R . Let V_L, V_R be isometries onto L, R and $\Pi_L = V_L V_L^*$, $\Pi_R = V_R V_R^*$. Since $(BP_j)^* = P_j B^*$ has range in L , $BP_j = BP_j \Pi_L$, and summing over j gives $B = B \Pi_L$; similarly $C = \Pi_R C$. Hence

$$BD_x W D_y C = (BV_L) \left(\sum_j x_j V_L^* P_j V_L \right) (V_L^* W V_R) \left(\sum_k y_k V_R^* P_k V_R \right) (V_R^* C).$$

In an orthonormal basis of L adapted to its decomposition, $\sum_j x_j V_L^* P_j V_L$ is diagonal with each x_j repeated $\text{rank}(P_j|_L)$ times; it is a unimodular diagonal, and likewise on R . The compressed middle $V_L^* W V_R$ is a contraction, and $\|BV_L\|_F^2 = \text{Tr}(B^* B \Pi_L) = \|B\|_F^2$, $\|V_R^* C\|_F = \|C\|_F$. Theorem 4.1 applies to the compressed architecture: its statement has a square middle layer, but its proof never uses squareness (Remark 4.4), and with a $\dim L \times \dim R$ middle the Khintchine step gives $\sqrt{\log(\dim L + \dim R)} \leq \sqrt{\log(2MN)}$. The supremum over all unimodular diagonals on L and R dominates the supremum over the repeated patterns. Corollary 5.1 then reads $\log \text{Pack} \leq \kappa \rho^{-2} N \log(2MN)$, and the $\Omega(N^2)$ demand of a Haar-scale code, with the error tolerance of Corollary 5.2, forces $\log M = \Omega(N)$. $\square$

For a bit-flip oracle $O_f|x, b\rangle = |x, b \oplus f(x)\rangle$ the eigen-decomposition of the response bit gives $M + 1$ addresses; nothing changes. The two selectors may also use different address decompositions, as for two tables of lengths M_1 and M_2 : the proof is unchanged with $\dim L \leq M_1 N$ and $\dim R \leq M_2 N$. The factorization, the dimension count, the preserved norms and the contractivity are checked on eighty random architectures with address projectors of arbitrary rank, and with two different decompositions, in `verify_address_compression.py`.

6.3 The matching upper bound

Theorem 6.4 (permutation transport). *For every $\varepsilon \in (0, \frac{1}{10})$ there are a target-independent encoder and decoder such that every $U \in U(N)$ is implemented by a clean two-query circuit with*

<i>resource</i>	<i>cost</i>
<i>Boolean tables</i>	<i>one</i>
<i>Boolean queries</i>	<i>two</i>
<i>oracle input length</i>	$O(N \log(1/\varepsilon))$ <i>bits</i>
<i>workspace</i>	$O(N \log(1/\varepsilon))$ <i>qubits</i>
<i>ordinary gates</i>	$\text{poly}(N, 1/\varepsilon)$
<i>half-diamond error, all reference-entangled inputs</i>	<i>at most ε</i>
<i>extra target-dependent advice, postselection</i>	<i>none</i>

The construction restores its workspace to within the stated error, so it is clean; the clean-output compression T_g is within operator-norm distance $3\varepsilon/4$ of U ; and the circuit is of the repeated-sign-word form T_g of Definition 2.1, one ± 1 table serving both queries.

Construction, condensed. The full statement with all estimates is in the Appendix A, also available as `notes/permutation-transport.md` in the repository. `verify_koopman_transport.py` checks the ingredients at small N .

Encode a state as a function on a finite grid rather than as a list of columns. Quantize the standard normal distribution into J equiprobable intervals with conditional means q_a , put $s^2 = \frac{1}{J} \sum_a q_a^2$ and $e^2 = 1 - s^2$; a direct estimate gives $e^2 \leq 32J^{-1/3}$ (natural logarithms throughout), so $J = \text{poly}(1/\varepsilon)$ makes e as small as needed, independently of N . For addresses

$a \in [J]^{2N}$, $M = J^{2N}$, define $w_a \in \mathbb{C}^N$ by $(w_a)_k = (q_{a_k} + iq_{a_{N+k}})/(s\sqrt{2})$ and the fixed encoding $(E\psi)_a = w_a^* \psi / \sqrt{M}$. Independence, zero means and the normalization give $E^*E = I_N$ exactly.

For a target U with realification O_U , draw $X \sim \mathcal{N}(0, I_{2N})$ and quantize coordinatewise with the cell map θ : $\alpha = \theta(X)$, $\beta = \theta(O_U X)$. Both are uniform on $[J]^{2N}$, so $\Xi_{ba} = M \Pr(\alpha = a, \beta = b)$ is doubly stochastic and, by Birkhoff–von Neumann [Bir46], a convex combination $\sum_{\ell=1}^m p_\ell P_{\pi_\ell}$ of at most $m \leq M^2$ permutation matrices. The quantization residual $r(X) = X - q_{\theta(X)}$ has covariance $e^2 I_{2N}$, and $q_{\theta(O_U X)} - O_U q_{\theta(X)} = O_U r(X) - r(O_U X)$, whence, using $(a - b)(a - b)^\top \preceq 2aa^\top + 2bb^\top$,

$$\sum_{\ell} p_{\ell} \Delta_{\pi_{\ell}}^* \Delta_{\pi_{\ell}} \preceq \frac{4e^2}{s^2} I_N, \quad \Delta_{\pi} := P_{\pi} E - E U.$$

This is a uniform operator bound, valid for every input. To make the mixture a single permutation, put $\delta := \varepsilon/2$, take J with $e^2 \leq \delta^2/100$, and add a uniform coherent label register over $R \geq 100M^2/\delta^2$ copies, the least power of two satisfying this bound, fixed by M and δ alone and hence independent of the target, with integer multiplicities $k_{\ell} \approx R p_{\ell}$, $\sum_{\ell} |p_{\ell} - k_{\ell}/R| \leq 2m/R$; set $\Pi_U(j, a) = (j, \pi_{\ell(j)}(a))$ with the enlarged, still target-independent, encoding $\tilde{E}\psi = |+_R\rangle \otimes E\psi$. Then

$$\|\Pi_U \tilde{E} - \tilde{E} U\|_{\text{op}}^2 = \left\| \sum_{\ell} \frac{k_{\ell}}{R} \Delta_{\pi_{\ell}}^* \Delta_{\pi_{\ell}} \right\|_{\text{op}} \leq \frac{4e^2}{s^2} + \frac{8m}{R} < \delta^2,$$

so the label register returns to $|+_R\rangle$ up to δ , uncorrelated with the input. The permutation Π_U and its inverse are stored in one Boolean table on $1 + 2t$ input bits, $t = \log_2(MR) = O(N \log(1/\varepsilon))$, and are applied by two queries in the standard way (load $\Pi_U(z)$ into a blank register, swap, unload with the inverse table). The fixed encoder is an explicit polynomial-size circuit and its inverse decodes. Total isometry error at most $3\varepsilon/4$: $\delta = \varepsilon/2$ from the label register and $\varepsilon/8$ each for the encoder and the decoder. Two isometries at operator distance γ induce channels at half-diamond distance at most γ . The full estimates are in the note.

The lower bound of Theorem 6.3 shows that for this mechanism, and indeed for any monomial action between fixed encodings, the exponential address length is unavoidable; the note records that converse and its one loophole, a target-dependent final ancilla state, which does not affect the clean model treated here.

Corollary 6.5 (the thresholds). *For clean two-query synthesis of all n -qubit unitaries at operator-norm error ε below an absolute constant, both the qubit count w^* and the oracle input length ℓ^* satisfy*

$$\Omega(2^n) \leq w^*, \ell^* \leq O(2^n \log(1/\varepsilon)).$$

7 The garbage model: what the two-query problem costs there

In the garbage model the workspace may end in a target-dependent state $|j_g\rangle$, independent of the input, or may simply be discarded. [LMW24] and [DLM26] state their one-query bounds there. This section gives a sufficient route from an insertion-class capacity bound to a lower bound in that model.

Setting. A t -query circuit $V(g) = A_t(D_g \otimes I_k)A_{t-1} \cdots (D_g \otimes I_k)A_0$ on $\mathbb{C}^Q \otimes \mathbb{C}^k$, where Q is the number of addresses and k the workspace dimension, restricted to the clean input, $S_g := V(g)\iota$ with $\iota\psi = \psi \otimes |0\rangle$, is an isometry $\mathbb{C}^N \rightarrow \mathbb{C}^Q \otimes \mathbb{C}^k \cong \mathbb{C}^N \otimes \mathbb{C}^D$, with $D = Qk/N$

the junk dimension, whose entries are multilinear polynomials of degree at most t in the signs. In the **keep form** $S_g\psi = (U_g\psi) \otimes |j_g\rangle$. The **discard form** only requires the induced channel to be within diamond distance ε of U_g ; since a unitary channel has Kraus rank one, continuity of the Stinespring representation recovers the keep form up to $O(\sqrt{\varepsilon})$ in operator norm: [KSW08] gives the required unitary on a common enlarged environment, which suffices here because embedding the junk register isometrically in a larger one changes none of the conjugated objects below, and [vE23] gives it on the junk register itself with constant $\sqrt{2}$. So everything below transfers to the discard form at constant error.

Lemma 7.1 (junk-blindness). *Call a functional L on isometries junk-blind if $L(S) = L((I \otimes W)S)$ for every unitary W on the junk register, equivalently if it depends only on the induced channel. If L is junk-blind and linear then $L \equiv 0$.*

Proof. Take $W = e^{i\theta}I$: junk-blindness gives $L(S) = L(e^{i\theta}S) = e^{i\theta}L(S)$ for all θ ; for a real-linear L take $\theta = \pi$. $\square$

The lemma rules out a nonzero junk-blind linear functional of the packet itself. Conjugation provides a useful quadratic invariant with an explicit $2t$ -insertion representation. This proves sufficiency of the transfer below, not necessity of oracle degree $2t$: substitution of an oracle circuit into an invariant can cause degree cancellations. For example, $D_g D_g = I$ is a two-query packet with constant conjugated observables. No lower bound on the degree of every possible invariant, or on every width method, is asserted.

Theorem 7.2 (conjugation transfer). *For every fixed compatible tuple of contractions $(Y_0, \dots, Y_{2t})$, consider the $N \times N$ family $\{Y_{2t}D_g Y_{2t-1} \cdots D_g Y_0 : g \in \{\pm 1\}^Q\}$ with $D_g = \sum_{j \leq Q} g_j P_j$ for mutually orthogonal projectors P_j of arbitrary rank summing to the identity of a register of arbitrary dimension, as in Theorem 6.3. Suppose each such family satisfies a capacity bound $\log \text{Pack}_{\rho\sqrt{N}} \leq C_\rho N \phi_N(Q)$, uniformly over the fixed contractions, projectors and workspace dimension. Packing is over g alone, not over these fixed architectural choices. Here ϕ_N is a rate function of the address count, allowed to depend on N but not on workspace dimension. Then no universal garbage- t synthesizer exists with $\phi_N(Q) = o(N)$, for every workspace and junk dimension, with no change to the oracle, at sufficiently small constant diamond error. At $t = 1$, Theorem 6.3 gives $\phi_N(Q) = \log(2QN)$.*

Proof. Fix a balanced reflection R ($R = R^*$, $R^2 = I$, $\text{Tr } R = 0$) and set $T_g := S_g^*(R \otimes I)S_g$. Because $D_g \otimes I$ is Hermitian, expanding S_g^* by reversal writes T_g exactly as a $2t$ -insertion product of the same sign word on $\mathbb{C}^Q \otimes \mathbb{C}^k$, with address projectors $P_j \otimes I_k$ and contractions in every slot (the middle slot is $A_t^*(R \otimes I)A_t$; the product is palindromic, $T_g = Y_g^* A_t^*(R \otimes I)A_t Y_g$ with $Y_g = (D_g \otimes I)A_{t-1} \cdots (D_g \otimes I)A_0$). In the keep form $T_g = \langle j_g | j_g \rangle U_g^* R U_g = U_g^* R U_g$: the junk cancels by the normalization of a unit vector, with no reference vector, no alignment and no dependence on the junk dimension. The map $U \mapsto U^* R U$ lands in the balanced reflections, a copy of the Grassmannian $\text{Gr}(N/2, N)$ of real dimension $N^2/2$ on which every point has Frobenius norm $\sqrt{N}$. Here is a direct packing justification: for Haar V , $f(V) = \text{Tr}(RV^*RV)$ is real, has mean zero, and is $2\sqrt{N}$ -Lipschitz in Frobenius distance. A ball of radius $\rho\sqrt{N}$ about R is the event $f(V) \geq N(1 - \rho^2/2)$. The same concentration inequality as in Corollary 5.2 bounds its measure by $\exp(-c_\rho N^2)$ for fixed $\rho < \sqrt{2}$. Invariance gives the same measure at every orbit point; a maximal separated set covers the orbit and therefore has at least $e^{c_\rho N^2}$ points, each of the form $U_i^* R U_i$. A universal synthesizer must synthesize $U_1, \dots, U_m$, so the single family $\{T_g\}$ contains an $e^{c_\rho N^2}$ -point separated code, contradicting the assumed capacity bound when $\phi_N(Q) = o(N)$. For the discard form no Stinespring continuity is needed: if $\|\Phi_g - \text{Ad}_U\|_\diamond \leq \varepsilon$, channel duality gives $\|\Phi_g^*(R) - U^* R U\|_{\text{op}} \leq \varepsilon$. Thus the normalized Frobenius error is at most ε , absorbed by taking a sufficiently small constant service error. $\square$

The reversal identity, the junk cancellation and the orbit properties are checked in `verify_conjugation_transfer.py`.

Corollary 7.3 (one query, garbage model). *At $t = 1$ the hypothesis of Theorem 7.2 is Theorem 6.3, applied to two insertions of a repeated real sign word. Hence no one-query circuit, with garbage allowed and with any number of ancillas, synthesizes every unitary at constant diamond error unless its oracle input length is $\Omega(2^n)$.*

This recovers, by a different route and at constant error only, the address-length form of the one-query bound of [LMW24]; we claim no improvement on their statement, which also rules out vanishing advantage.

Remark 7.4 (what is not claimed). At $t = 2$ the hypothesis of Theorem 7.2 is a clean **four**-insertion width bound, which is open. By the palindromic form it suffices to bound the sandwich subclass $Y_g^*XY_g$ with Y_g a two-insertion packet whose left endpoint is the identity of the full register: exactly the case in which the endpoint budget that makes step 3 of Theorem 4.1 work is unavailable, since that identity has squared Frobenius norm equal to the register dimension. So the unrestricted garbage-model two-query problem is open, and this paper does not claim it. A restricted two-query garbage bound, valid when the junk register has fewer than $2n - O(1)$ qubits, exists in the author’s working records but has not passed an independent audit and is not included.

8 Scope, and depth three and beyond

What Theorem 4.1 covers. Exact and approximate clean two-query synthesis; arbitrary contractive B, W, C ; real sign oracles, diagonal-unitary oracles and two different tables; ancillas explicitly inside $\log(2Q)$, and, via Theorem 6.3, the oracle input length alone.

What it does not cover.

1. **Three or more queries.** Nothing here bears on $t \geq 3$. The natural conjecture, that the width of the k -query clean class is $O(N \cdot k \cdot \text{polylog } Q)$, is pinned in exponent by [Ros26]: every unitary is implementable at $k = \tilde{O}(\sqrt{N})$ with $\log Q = \text{poly}(n)$, so the width must reach the $\Theta(N^{3/2})$ width of the unitary group there, which forces at least linear growth in k . A proof of the conjecture would resolve the Aaronson–Kuperberg question negatively. Less would do: any bound of the form $N \cdot \text{poly}(k, \log Q)$ already excludes universal clean synthesis at polynomial depth and polynomial qubit count, since such a class could not ε -cover $U(N)$, whose width is $\Theta(N^{3/2})$ (the argument of Corollary 6.2); and by Theorem 7.2 the same bound at $2k$ insertions, in address form, would exclude it in the garbage model as well. The two-query proof does not extend as it stands: with a third insertion the coefficient matrix depends on the selector.
2. **Parallel versus adaptive.** [LMW24] rule out polynomially many *parallel* queries. The difficulty at $t \geq 3$ is therefore entirely in interleaving.
3. **One query.** [AK07, Thm 6.7] is stronger at $t = 1$ in the exact model: 4^N with at most $2N$ relevant table entries. This counts Boolean variables, not oracle input bits. Banerjee [Ban25] sharpens the matrix count to 2^N under the exact every-oracle-clean assumption.
4. **Completely bounded forms.** [ABP18, Thm 1.3] characterizes scalar expected outputs of t -query algorithms by completely bounded forms of degree $2t$. Our matrix-valued amplitude packet is separately multilinear of degree t . These are related factorizations, but the cited scalar theorem is not an equivalence with the full matrix-valued unit ball. Such an identification is not used or claimed here.

Consistency checks. The bound does not forbid the trivial clean two-query lookup ($w = \Theta(N^2)$), is attained up to $\log(1/\varepsilon)$ by Theorem 6.4, and does not contradict [Ros26], whose algorithm uses $t = O(\sqrt{N})$ queries. Any extension of this method yielding $t = \omega(\sqrt{N})$ in the clean model would contradict [Ros26] and be in error.

9 Reproducibility

`verify/run_all.py` in the repository runs nine deterministic scripts (standard library and numpy), each exiting nonzero on any failed check: Theorem 4.1 in both forms (exact expansion, exhaustive absorption on finite sign cubes and checks at sampled or alternating-maximization phase vectors, the closed-form variance operators against Monte Carlo and their domination), Proposition 2.2, Propositions 3.1 and 3.2, Proposition 5.3, Proposition 6.1, the factorization of Theorem 6.3, the ingredients of Theorem 6.4 at small N , and the identities of Section 7. The `lean/` directory holds a kernel-checked formalization of the deterministic spine of Corollary 4.2, with the analytic inputs as explicit hypotheses; its `README` states exactly what is and is not certified.

Acknowledgements and disclosure

Use of language models. Large language models were used extensively throughout this work for proof discovery, implementation, verification engineering, adversarial auditing, and editorial assistance. The workflow used separate, independently instantiated sessions, including fresh-context auditing sessions instructed to refute rather than confirm. The author directed the research program, set the verification standards, made the promotion, correction, and publication decisions, and assumes sole responsibility for this work. These AI audits are not external peer review.

The proofs are analytic, model-assisted arguments; exact computation and repeated audits do not by themselves validate their analytic dependencies. The scripts check specified identities and finite instances, using exact rational arithmetic for designated controls and floating-point calculations elsewhere; numerical searches do not certify global optima. The Lean kernel checks seven selected declarations, including conditional assembly results whose analytic inputs remain explicit hypotheses. The complete width, packing, address-compression, and transport proofs are not formalized in Lean.

A Complete permutation-transport construction

This appendix proves Theorem 6.4 in the fixed clean-isometry metric, including the finite encoding, rounding, circuit implementation and all reference-entangled inputs. No efficient algorithm for constructing the target-dependent Boolean truth table is assumed or claimed.

A.1 Quantization with dimension-independent error

Let $J \geq 4$ be a power of two. Partition a real standard normal variable X into J equiprobable intervals, with cell index $\theta(X)$ and conditional means $q_a = \mathbb{E}[X \mid \theta(X) = a]$. Set

$$s^2 = J^{-1} \sum_a q_a^2, \quad e^2 = \mathbb{E}(X - q_{\theta(X)})^2 = 1 - s^2.$$

The means sum to zero. For completeness, $e^2 \leq 32J^{-1/3}$: write $g(u) = \Phi^{-1}(u)$ and clip it to $[-T, T]$ to obtain g_T , which is Lipschitz with constant $L_T = \sqrt{2\pi}e^{T^2/2}$. Orthogonal

projection onto cell-constant functions is the best L^2 approximation, so $e \leq \|g - g_T\|_2 + L_T/J$. For $T \geq 1$,

$$\|g - g_T\|_2^2 \leq \mathbb{E}[X^2 \mathbf{1}_{|X| > T}] = 2(T\phi(T) + 1 - \Phi(T)) \leq 2(T+1)\phi(T).$$

With $T = \sqrt{\log J}$ this gives

$$e^2 \leq \frac{4(\sqrt{\log J} + 1)}{\sqrt{2\pi J}} + \frac{4\pi}{J} \leq 32J^{-1/3}.$$

One elementary check of the last bound is to multiply by $J^{1/3}$: $(\sqrt{u}+1)e^{-u/6} \leq \sqrt{3}e^{-1/2}+1 < 3$ for $u \geq 0$ (maximize $\sqrt{u}e^{-u/6}$ at $u = 3$), and $4\pi J^{-2/3} \leq 4\pi/4^{2/3} < 5$. Thus the displayed constant is conservative.

A.2 A fixed finite isometry and its circuit

Put $\Omega = [J]^{2N}$, $M = J^{2N}$, and

$$(w_a)_k = \frac{q_{a_k} + iq_{a_{N+k}}}{s\sqrt{2}}, \quad (E\psi)_a = \frac{w_a^* \psi}{\sqrt{M}}.$$

Independence and zero means give $M^{-1} \sum_a w_a w_a^* = I_N$, so $E^*E = I_N$ exactly. Define orthonormal coordinate states

$$|g\rangle = J^{-1/2} \sum_a |a\rangle, \quad |h\rangle = (s\sqrt{J})^{-1} \sum_a q_a |a\rangle.$$

Then $E|j\rangle$ is the superposition of the configuration with h in coordinate j and the one with h in coordinate $N+j$, with coefficients $1/\sqrt{2}$ and $-i/\sqrt{2}$; all other coordinates hold g . Coherently convert the binary label j into these two possible single excitation positions, and erase the binary label by computing it back from the occupied position. Apply on each coordinate a fixed unitary taking $|0\rangle$ to g and $|1\rangle$ to h . This gives an explicit unitary extension of E with $O(N(n+1)^2 + NJ^2 \text{poly}(\log J, \log(N/\varepsilon)))$ gates after finite-precision compilation. Compile the whole encoder to operator error $\varepsilon/8$ and use its circuit inverse for decoding. Every angle is target independent. This gate count is polynomial in $N, 1/\varepsilon$, not in n .

A.3 Transport coupling and its operator error

For a fixed target U , let

$$O_U = \begin{pmatrix} \Re U & -\Im U \\ \Im U & \Re U \end{pmatrix}.$$

Take X standard normal in $\mathbb{R}^{2N}$ and set $\alpha = \theta(X)$, $\beta = \theta(O_U X)$ coordinatewise. Both marginals are uniform on Ω , so $\Xi_{ba} = M \Pr(\alpha = a, \beta = b)$ is doubly stochastic. It is a convex combination $\sum_{\ell=1}^m p_\ell P_{\pi_\ell}$ with $m \leq M^2$. To see finiteness directly, Hall's condition follows from equal row and column sums on the positive support; subtract the smallest matching entry at each step. Row and column sums stay equal, at least one positive entry disappears, and the process ends in at most M^2 steps. The weights sum to one.

Write $r(X) = X - q_{\theta(X)}$. Its coordinates are independent, centered, and have variance e^2 , and the same distributional statement holds for $r(O_U X)$. Therefore

$$\mathbb{E}[(q_\beta - O_U q_\alpha)(q_\beta - O_U q_\alpha)^\top] \preceq 4e^2 I_{2N}.$$

Here we used $q_\beta - O_U q_\alpha = O_U r(X) - r(O_U X)$ and $(a - b)(a - b)^\top \preceq 2aa^\top + 2bb^\top$; no independence between these two residual vectors is needed. The map $(x, y) \mapsto (x + iy)/(s\sqrt{2})$ has squared operator norm $1/s^2$, so

$$C_U := \mathbb{E}[(w_\beta - U w_\alpha)(w_\beta - U w_\alpha)^*] \preceq (4e^2/s^2)I_N.$$

For $\Delta_\pi = P_\pi E - EU$, expansion at output coordinate $b = \pi(a)$ gives, for every ψ ,

$$\sum_\ell p_\ell \|\Delta_{\pi_\ell} \psi\|^2 = \mathbb{E}|w_\alpha^* \psi - w_\beta^* U \psi|^2 = \psi^* U^* C_U U \psi.$$

Consequently

$$\sum_\ell p_\ell \Delta_{\pi_\ell}^* \Delta_{\pi_\ell} \preceq (4e^2/s^2)I_N. \quad (1)$$

This bounds all inputs simultaneously, not only the maximally mixed input.

A.4 One coherent permutation and a fixed label state

Set $\delta = \varepsilon/2$ and choose J with $e^2 \leq \delta^2/100$. Choose the least power of two $R \geq 100M^2/\delta^2$ depending only on M, δ . Round the weights to integers $k_\ell \geq 0$ with $\sum k_\ell = R$ and $\sum |p_\ell - k_\ell/R| \leq 2m/R$: round down first, then distribute the remaining slots. List k_ℓ copies of each permutation and define

$$\Pi_U(j, a) = (j, \pi_{\ell(j)}(a)), \quad \tilde{E}\psi = |+_R\rangle \otimes E\psi.$$

Orthogonality of the label states gives the exact identity

$$\|\Pi_U \tilde{E} - \tilde{E}U\|_{\text{op}}^2 = \left\| \sum_\ell (k_\ell/R) \Delta_{\pi_\ell}^* \Delta_{\pi_\ell} \right\|_{\text{op}}.$$

Since $\|\Delta_\pi\| \leq 2$, (1) implies

$$\|\Pi_U \tilde{E} - \tilde{E}U\|_{\text{op}}^2 \leq 4e^2/s^2 + 8m/R < \delta^2.$$

Thus the label returns approximately to the same uniform state; it is not measured, discarded, or allowed to retain target-dependent junk. The binary coordinate and label registers have $t = \log_2(MR) = O(N \log(1/\varepsilon))$ qubits in total.

A.5 Two Boolean queries, complete address count, and error

On a direction bit d and two t -bit strings z, y , define one table

$$f_U(d, z, y) = \begin{cases} \langle \Pi_U(z), y \rangle_{\mathbb{F}_2}, & d = 0, \\ \langle \Pi_U^{-1}(z), y \rangle_{\mathbb{F}_2}, & d = 1. \end{cases}$$

The full oracle input length is $1 + 2t$. A phase query sandwiched between Hadamards on y implements XOR lookup of the entire t -bit answer, by character orthogonality. Forward lookup, register swap, and inverse lookup take $|z, 0\rangle$ to $|\Pi_U(z), 0\rangle$. The direction bit and the phase-answer qubit are restored by fixed gates. Both lookups query the same table, in two disjoint tagged sections. The remaining gates and encoder/decoder are fixed before U is chosen.

Decoding the transport error and including both compilation errors gives full-isometry distance at most $\delta + 2(\varepsilon/8) = 3\varepsilon/4$ from $\psi \mapsto U\psi \otimes |0 \cdots 0\rangle$. Tensoring with a reference preserves this operator bound. The half trace distance between the corresponding pure states is at most their vector distance; partial trace and convexity extend the guarantee to arbitrary mixed reference-entangled inputs. Thus the half-diamond error is at most ε , with approximate cleanup in the stronger full-isometry sense. No bound on the offline time or size of the truth table is claimed.

A.6 A converse for monomial action between fixed encodings

Let $E, F : \mathbb{C}^N \rightarrow \mathbb{C}^D$ be fixed isometries and let the only target-dependent operation be a monomial P . Write $e_i = E^*|i\rangle$, $f_j = F^*|j\rangle$, and define

$$a(U) = \max_{i,j:e_i, f_j \neq 0} \frac{|\langle U e_i, f_j \rangle|}{\|e_i\| \|f_j\|}.$$

For every permutation and every choice of its phases, $|\text{Tr}(U^* F^* P E)| \leq N a(U)$ by Cauchy–Schwarz and $\sum \|e_i\|^2 = \sum \|f_j\|^2 = N$. If the two isometric channels are within half-diamond error ε , their pure normalized Choi states have squared overlap at least $1 - \varepsilon^2$, forcing $a(U)^2 \geq 1 - \varepsilon^2$. For fixed unit vectors v, w and Haar U , $\Pr(|\langle U v, w \rangle|^2 \geq 1 - \varepsilon^2) = \varepsilon^{2(N-1)}$. A union bound over at most D^2 pairs gives universality only if $D^2 \varepsilon^{2(N-1)} \geq 1$, or $\log_2 D \geq (N-1) \log_2(1/\varepsilon)$. This is a converse for fixed output encoding including its ancilla, not for a target-dependent final garbage state.

B Exact finite control for Proposition 3.2

This example counts scalar unitary matrices, not distinct channels: at $N = 1$ all these matrices induce the same channel. It is not used in any lower bound. Index the eight coordinates by $x \in \mathbb{F}_2^3$ in lexicographic order. Write $\chi_a(x) = (-1)^{a \cdot x}$, $A(a_0, a_1, a_2) = (a_2, a_0 + a_2, a_1)$ and $f = 8^{-1/2}(1, \dots, 1)$. Both A and $I + A$ are invertible over $\mathbb{F}_2$. Take $C = f$, $B = f^*$ and define

$$W_1 = \frac{1}{8} \sum_a \chi_{Aa} \chi_a^\top, \quad W_2(\lambda) = \frac{1}{8} \sum_a \lambda_a \chi_a \chi_{(I+A)a}^\top.$$

The character bases are orthogonal, so both mixers are unitary when $|\lambda_a| = 1$. For $g = \pm \chi_a$, direct substitution gives $T_g = \pm \lambda_a$ and no leakage. Use $\lambda_a = 1$ for the first architecture. For the second, enumerate a by $j = 0, \dots, 7$ and set $\lambda_a = (1 - j^2 + 2ij)/(1 + j^2)$. The sixteen values $\pm \lambda_a$ are distinct. For completeness, the exact finite certificate in `verify/verify_exact_controls.py` enumerates all 256 sign words, forms $v = D_g W_1 g$, and tests whether all entries of $D_g W_2 v$ are identical. In rational real and imaginary arithmetic the accepted words are precisely the sixteen $\pm \chi_a$ for both architectures. This is an exhaustive finite certificate, with no floating-point tolerance. The frames before W_2 agree because C, W_1 agree.

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